Note

There is one more criterion to check the congruency of triangle. This criterion is called Angle– Angle – Side criterion which is a slight modification of ASA criterion. In this criterion the congruent side is not included between the congruent angles. So we can say, if two angles and a non-included side of one triangle are congruent to the corresponding parts of another triangle, then also the triangles are congruent.

Hypotenuse

In earlier class we learnt about right angled triangle. In a right angled triangle, one angle is a right angle and other angles are acute angles.

Observe the following right angled triangles.

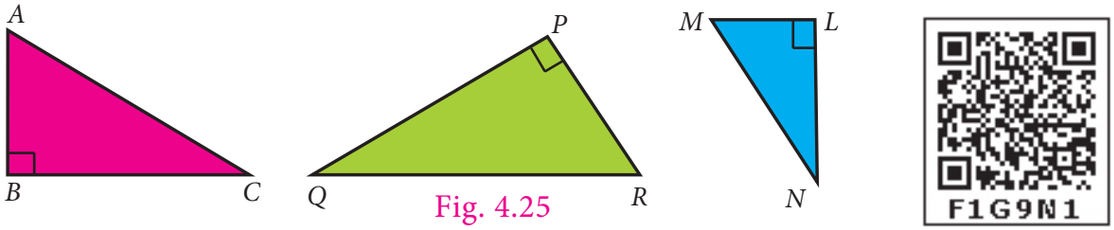


Fig. 4.25

In all the given triangles the side which is opposite to the right angle is the largest side called Hypotenuse.

In the above triangles the sides *AC*, *QR* and *MN* are the hypotenuse. Remember that hypotenuse is related with right angled triangles only.

4. Right Angle – Hypotenuse – Side congruence criterion (RHS)

The name of the criterion clearly states that this criterion is used with right angled triangles only.

Observe the two given right angled triangles.

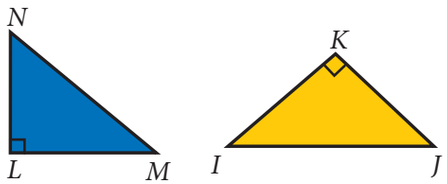


Fig. 4.26

In these two triangles right angle is common. And if we are given the sides making right angles then we can use the *SAS* criterion to check the congruency of the triangles.

Or if we are given one side containing right angle and hypotenuse, then we can have new criterion, if the hypotenuse and one side of a right angled triangle is equal to the hypotenuse and one side of another right angled triangle then the two right angled triangles are congruent.

This is called Right angle – Hypotenuse – Side criterion.

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